

Digital Enablement for ESG Compliance in Manufacturing

Thought Leadership Whitepaper | Sustainability Practice

Abstract

ESG reporting is rapidly transitioning from voluntary to mandatory for large Indian manufacturers, driven by SEBI's BRSR framework, OEM supply-chain requirements, and international investor expectations. This whitepaper examines how digital tools — IoT energy monitoring, waste analytics, and integrated compliance dashboards — enable manufacturers to move from ad-hoc ESG reporting to real-time, auditable sustainability metrics.

1. The ESG Reporting Imperative

India's Business Responsibility and Sustainability Report (BRSR) requirement now applies to the top 1,000 listed companies by market capitalisation. Supply chain ESG expectations are cascading further: automotive and FMCG OEMs are requiring Tier-1 and Tier-2 suppliers to report Scope 3 emissions data, energy intensity, and water usage. Manufacturers that cannot provide this data risk losing contracts.

2. Key ESG Metrics in Manufacturing

Energy & Carbon:

- Energy intensity (GJ per tonne of production) — Scope 1 & 2 emissions tracking
- Renewable energy percentage — solar, wind, and green tariff procurement
- Carbon footprint tracking aligned to GHG Protocol and ISO 14064

Waste & Water:

- Solid waste generation and recycling rates by category
- Effluent treatment performance: BOD, COD, heavy metals against consent limits
- Water intensity (kL per unit of production) and water recycling percentage

3. Digital Tools for ESG Enablement

- IoT sub-metering: automatic, granular energy data by asset, line, and building
- Waste tracking systems: digital records replacing manual logbooks for CPCB/SPCB compliance
- EHS Management Systems: incident tracking, compliance calendar, permit management
- ESG Dashboards: real-time BRSR KPIs, audit-ready data export, trend analysis
- Carbon accounting software: automatic Scope 1, 2, 3 calculation from operational data

4. Suggested Use in Sales Conversations

Use this whitepaper to open ESG-adjacent conversations with CFOs and sustainability heads at large manufacturers. Key conversation hooks: (1) BRSR compliance deadline pressure — many clients do not yet have auditable digital records; (2) OEM supply-chain ESG requirements — this is an immediate revenue risk for Tier-1 and Tier-2 suppliers; (3) Cost reduction framing — energy

monitoring pays for itself through consumption reduction before ESG compliance value is even counted.

5. Our ESG Digital Capability

- Carbon footprint baseline assessments completed for 14 manufacturing clients since 2021
- IoT-based energy monitoring deployed at 8 facilities, average energy saving of 16%
- ESG compliance dashboards live at 6 clients across automotive, FMCG, and process industries
- Partnerships with leading carbon accounting platforms for data integration
- ISO 14001 and ISO 50001 implementation support available as an add-on service